Chapter 3 Magnetostatics

Unlike the electric field, the magnetic field (B field) exhibits remarkably strange behavior. However, studying the B field becomes more intuitive when compared to the E field. One can observe a strikingly parallel set of physical laws governing both fields. In this chapter, we will take this approach—exploring magnetostatics while revisiting electrostatics.

1. The static magnetic field

1) Review of static *E* field

The sources generating the electrostatic field are static charges. We can derive everything about electrostatics starting from two basic laws:

$$\text{Law 1: Coulomb's law}: \ \vec{F} = \underbrace{\frac{1}{4\pi\varepsilon_0}\frac{q}{\mathscr{r}^2}\cdot\hat{\mathscr{r}}}_{\vec{E}\ field}\cdot Q = Q\cdot\vec{E}$$

$$\text{Law 2: superposition}: \ \vec{F}_{tot} = \sum_i \vec{F}_i$$

2) Review of static B field

Compared to the electrostatic field, the magnetostatic field is significantly more complex. This complexity is evident in its directional behavior, which follows the right-hand rule (see the figure below). We will explore the field in a more quantitative manner later.

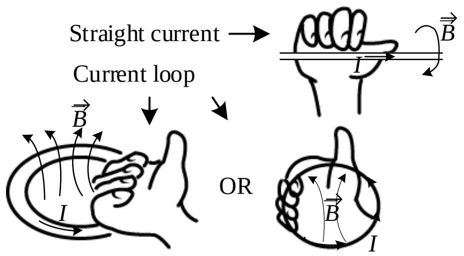

Figure. Use the right-hand rule to determine the direction of the B field.

The source-field relationship in magnetostatics is more complex than in electrostatics. Similarly, the interplay between the Lorentz force, the B field, and the test current (analogous to a test charge in electrostatics) follows the left-hand rule (see below).

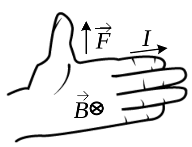

Figure. Use the left-hand rule to determine the direction of the Lorentz force.

3) The source of magnetostatic field

The magnetostatic field is generated by current, which consists of continuously moving charges. More precisely, current is defined as the amount of charge passing through a surface per unit time. Thus, unlike electrostatics, the source of the magnetostatic field is inherently dynamic.

One might assume, by analogy with electrostatics (where point charges are the fundamental source), that a moving charge generates a magnetostatic field and that the full field follows from superposition. However, strictly speaking, discrete moving charges cannot sustain a static magnetic field. Instead, a steady current is required, making the current element the fundamental source of magnetostatics. We will study what is a steady current soon.

Note that an electric wire carrying a steady current remains electrically neutral, as negative charges flow within a fixed matrix of an equal amount of positive charges, see the figure below.

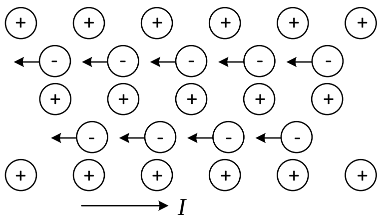

Figure. Zoomed-in view of a piece of electrically neutral wire carrying a current.

4) Steady current

① General property of currents

Currents represent the flow of charges. Similar to charge density, which describes the spatiotemporal distribution of charges, current density characterizes the spatiotemporal distribution of currents. When represented with arrows indicating local flow directions, current density resembles a vector field, as illustrated below.

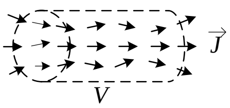

Figure. An illustration of current density field $\vec{J}$ within a volume V.

The definition of the current density is that $|\vec{J}|$ represents the number of charges per unit time per unit area flowing through a surface, where $\hat{J}$ is a unit vector pointing along the local flow direction. Given a surface with a current I flowing through it, $|\vec{J}| = \frac{I}{A}$, where A is the area.

Now, consider the volume V shown in the figure above, enclosed by the surface S. By charge conservation, over a short time interval dt, the increase in charge within V equals the total charge flowing into V through S. Mathematically,

$$\frac{dQ}{dt} = \frac{d(\int_V \rho dV)}{dt} = -\oint_S \vec{J} \cdot d\vec{a}.$$

By convention, $d\vec{a}$ points outward, so $-\oint_S \vec{J} \cdot d\vec{a}$ is the total charge flowing into the volume per unit time. Since we keep the boundary of the volume fixed, $\frac{d(\int_V \rho dV)}{dt} = \int_V \frac{\partial \rho}{\partial t} dV$, invoking divergence theorem, $\oint_S \vec{J} \cdot d\vec{a} = \int_V (\nabla \cdot \vec{J}) dV$, we reach the following formula

$$\int_V (\nabla \cdot \vec{J}) dV = -\int_V \frac{\partial \rho}{\partial t} dV$$

which holds for any volume. Consequently, we readily obtain

$$\boxed{\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}}$$

This formula, known as the continuity equation, expresses charge conservation in classical electrodynamics. It explicitly relates $\vec{J}$ and ρ, both of which depend on space (x, y, z) and time (t). The equation states that the source of $\vec{J}$, $\nabla \cdot \vec{J}$, represents the negative rate of change of the local charge density. According to the continuity equation, a diverging current density at position $\vec{r}$ indicates that the charge density at $\vec{r}$ is decreasing over time, see the following figure.

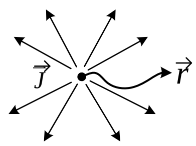

Figure. A pattern showing $\nabla \cdot \vec{J} > 0$ at $\vec{r}$. Apparently charges are leaving this point, resulting in a time-decreasing local charge density at $\vec{r}$.

② Steady current

In magnetostatics, a stable magnetic field is generated by a steady current, meaning both the current density distribution and the charge distribution must be time-independent. According to the continuity equation, these conditions translate to

$$\boxed{\frac{\partial \rho}{\partial t} = 0, \qquad \frac{\partial \vec{J}}{\partial t} = 0, \qquad \nabla \cdot \vec{J} = 0}$$

A steady current flows consistently, there is no accumulation or depletion of charge anywhere.

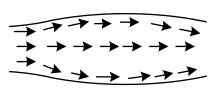

Figure. An illustration of a steady current.

5) The Lorentz force

The Lorentz force describes the force exerted on a charged particle moving in an electromagnetic field. The electric field, magnetic field, charge amount, and moving velocity are denoted as $\vec{E}$, $\vec{B}$, Q, $\vec{v}$, respectively. The force due to the $\vec{E}$ field is: $\vec{F_E} = Q\vec{E}$ (we've learnt this in electrostatics). The force due to the $\vec{B}$ field is $\vec{F_B} = Q(\vec{v} \times \vec{B})$.

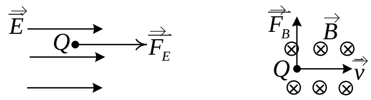

Figure. The electric force (left) and the magnetic force (right).

Note that the magnetic field is generated by moving charges and can only interact with other moving charges—static charges do not respond to magnetic fields. The Lorentz force acting on multiple moving charges, such as those in a current-carrying wire, can be determined using the superposition principle:

$$\vec{F_m} = \int (\vec{v} \times \vec{B}) dq.$$

For a thin wire with line charge density λ, $dq = \lambda dl$. $(\vec{v} \times \vec{B})dq = (\vec{v} \times \vec{B})\lambda dl$. Note that $\lambda\vec{v}$ is the total amount of charge flowing through the wire per unit time (v = dl/dt, $\lambda v = \frac{\lambda dl}{dt} = \frac{dq}{dt}$), which is the definition of a current: $\lambda\vec{v} = \vec{I}$. Here we write the current as a vector to specify its flowing direction. As a result, for a current-carrying wire, we can write:

$$\vec{F_m} = \int (\vec{I} \times \vec{B}) dl.$$

Moreover, charge can be distributed on a surface with a surface charge density of σ, moving of the surface charge gives rise to a surface current density, $\vec{K}$, which is the current <u>per unit width</u>. In this case, $(\vec{v} \times \vec{B})dq = (\vec{v} \times \vec{B})\sigma da$. Note that $\sigma = \frac{dq}{da}$, $v = \frac{dl}{dt}$, $da = dl \cdot dw$ (w denotes width), so $\sigma v = \frac{dq}{da}\frac{dl}{dt} = \frac{dq}{dw \cdot dt} = K$. Consequently, for the case of surface current,

$$\vec{F_m} = \int (\vec{K} \times \vec{B}) da.$$

In the most general case of volumetric charge distribution specified by ρ, movement of the charge generates a current density distribution of $\vec{J}$, which is the current per unit area. In this case, $(\vec{v} \times \vec{B})dq = (\vec{v} \times \vec{B})\rho dV$. Note that $\rho = \frac{dq}{dV}$, $v = \frac{dl}{dt}$, $dV = dl \cdot da$ (a denotes the crosssectional area), so $\rho v = \frac{dq}{dV}\frac{dl}{dt} = \frac{dq}{da \cdot dt} = \mathcal{J}$. Consequently, for the case of volume current,

$$\vec{F}_m = \int (\vec{\mathcal{J}} \times \vec{B}) dV.$$

From the above analysis, we conclude that, analogous to $dq \sim \lambda dl \sim \sigma da \sim \rho dV$ in electrostatics, the fundamental element in magnetostatics is $\vec{I} dl$ (which we may denote as $\vec{I} dl = I d\vec{l} = d\vec{I}$, representing a short segment of current) for a line current, $\sim \vec{K} da$ for a surface current, and $\sim \vec{\mathcal{J}} dV$ for a volume current.

Note:

Magnetic forces do no work, since it's always perpendicular to the moving direction:

$$(\vec{v} \times \vec{B}) \perp \vec{v} \Rightarrow (\vec{v} \times \vec{B}) \perp (\vec{v} dt) \Rightarrow (\vec{v} \times \vec{B}) \perp d\vec{l},$$

As a result, the work done by the magnetic force is

$$W_m = \int \vec{F}_m \cdot d\vec{l} = 0.$$

Example:

A rectangular current loop is attached to a weight m, with its upper edge of length l placed in a uniform magnetic field B. The directions of the current and magnetic field are shown in the figure below. An ideal, weightless battery maintains a constant current. If the current is strong enough, the magnetic force on the upper edge can exactly balance the gravitational force, allowing the system to move upward at a constant velocity. At first glance, it appears that the magnetic force is doing work, seemingly contradicting the principle that magnetic forces never do work. Explain this paradox.

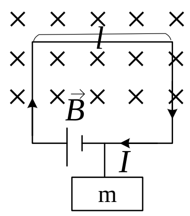

Figure. A current-carrying wire in a magnetic field can lift a weight.

Solution:

The key to resolving this paradox lies in recognizing that as the system moves upward, the charges in the upper edge of the wire do not move purely horizontally; instead, they acquire a vertical velocity due to the loop's upward motion. Now, we examine all the forces acting on a single moving charge, as illustrated in the following figure.

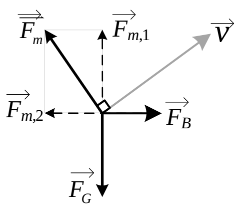

Figure. Analysis of the forces acting on a moving charge in the upper edge of the wire.

The magnetic force $\vec{F}_m$ is always perpendicular to the charge's velocity. It can be decomposed into two components: a vertical component $\vec{F}_{m,1}$ and a horizontal component $\vec{F}_{m,2}$. Since the charge moves at a constant speed, all forces acting on it must be balanced. The vertical component $\vec{F}_{m,1}$ is counteracted by gravity $\vec{F}_G$, while the horizontal component $\vec{F}_{m,2}$ is balanced by another horizontal force, $\vec{F}_B$, which is the electric force due to the battery. In this process, while $\vec{F}_{m,1}$ does positive work, $\vec{F}_{m,2}$ does negative work, resulting in a net zero work done by $\vec{F}_m$. Ultimately, the positive work is done by $\vec{F}_B$, i.e., the battery.

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6) The Biot-Savart Law

The Biot-Savart Law (sometimes called the Biot-Savart-Laplace Law), is the magnetostatic analogue of Coulomb's law in electrostatics. Since the entire framework of electrostatics is built upon Coulomb's law, the significance of the Biot-Savart Law becomes immediately apparent. To better understand the Biot-Savart Law, it is helpful to first revisit how Coulomb's Law is used to compute the electrostatic field. For a single charge, the electric field generated by the charge is determined by $\vec{E}_q(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{\mathscr{r}^2}\cdot\hat{\mathscr{r}}$, the parameters in the formular are shown in the figure below.

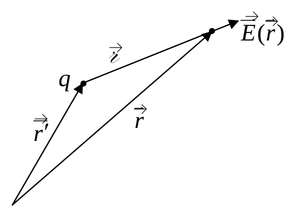

Figure. Using Coulomb's Law to determine the electric field.

For a continuous charge distribution, the principle of superposition is used to calculate the electric field:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\vec{r}')}{\mathscr{r}^2}\cdot\hat{\mathscr{r}}\,dV'.$$

The Biot-Savart Law for a "single current element" is

$$d\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{d\vec{I} \times \hat{\mathscr{r}}}{\mathscr{r}^2},$$

where $d\vec{I} = Id\vec{l}$ is the current element we discussed earlier, μ_0 is a constant called the permeability of vacuum, under SI unit, its value is $4\pi \times 10^{-7}$ H/m. The above law is illustrated in the following figure.

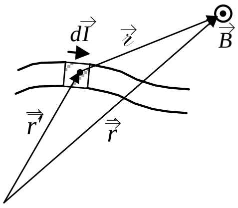

Figure. Using the Biot-Savart Law to determine the magnetic field.

Note that the equation above represents the elemental magnetic field $d\vec{B}$ generated by a current element $d\vec{I}$. While we drew an analogy to Coulomb's Law in electrostatics, an elemental magnetic field does not physically exist, as there is no such thing as an isolated current element. This contrasts with electrostatics, where a single charge element (a point charge) does exist in principle. The Biot-Savart Law becomes meaningful when applied through the superposition principle to calculate the magnetic field from a continuous current distribution. Recall that we learnt earlier $d\vec{I} = Id\vec{l}$ (line current) $= \vec{K}da$ (surface current) $= \vec{J}dV$ (volume current) , so in the case of a current-carrying wire (see the figure below, left),

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} I \int_L \frac{d\vec{l}' \times \hat{\mathscr{r}}}{\mathscr{r}^2},$$

while for a surface current distribution (see the figure below, middle),

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\int_S \frac{\vec{K}(\vec{r}') \times \hat{\mathscr{r}}}{\mathscr{r}^2} da',$$

And for a volume current distribution (see the figure below, right),

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \frac{\vec{J}(\vec{r}') \times \hat{\mathscr{r}}}{\mathscr{r}^2} dV'.$$

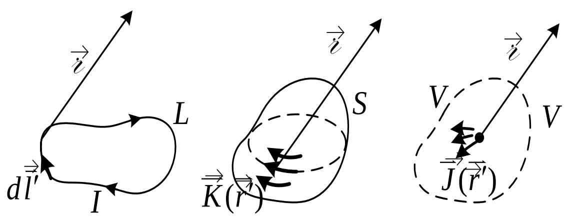

Figure. Three types of current distribution.

2. Properties of the magnetostatic field

In the previous Chapter, we derived the key properties of the electrostatic field – namely its divergence and curl – directly from Coulomb's Law. Here, we apply the same approach to determine the properties of the magnetostatic field by analyzing the Biot-Savart Law. We start from the most general form of volume current distribution:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \frac{\vec{J}(\vec{r}') \times \hat{\mathscr{r}}}{\mathscr{r}^2} dV' = \frac{\mu_0}{4\pi}\int_V \vec{J}(\vec{r}') \times \frac{\hat{\mathscr{r}}}{\mathscr{r}^2} dV'.$$

We can rewrite the above equation taking into account $\nabla\frac{1}{\mathscr{r}} = -\frac{\hat{\mathscr{r}}}{\mathscr{r}^2}$ to get

$$\vec{B}(\vec{r}) = -\frac{\mu_0}{4\pi}\int_v \vec{J}(\vec{r}') \times \nabla\frac{1}{\mathscr{r}} dV'$$

We apply the product rule $\vec{A} \times (\nabla f) = f(\nabla \times \vec{A}) - \nabla \times (f\vec{A})$ such that the integrand in the above equation becomes

$$\vec{J}(\vec{r}') \times \nabla\frac{1}{\mathscr{r}} = \underbrace{\frac{1}{\mathscr{r}}(\nabla \times \vec{J}(\vec{r}'))}_{\substack{\text{Note that } \nabla \text{ operates on } \vec{r},\ \text{not } \vec{r}' \\ \text{such that } \nabla\times\vec{J}(\vec{r}') = 0}} - \nabla \times \frac{\vec{J}(\vec{r}')}{\mathscr{r}}$$

Which leads to the following expression:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \nabla \times \frac{\vec{J}(\vec{r}')}{\mathscr{r}} dV' \xrightarrow{\nabla \text{ acts on } \vec{r},\ \text{integration is over } \vec{r}'} \nabla \times \left[\frac{\mu_0}{4\pi}\int_V \frac{\vec{J}(\vec{r}')}{\mathscr{r}} dV'\right]$$

$$\boxed{\vec{B}(\vec{r}) = \nabla \times \vec{A}(\vec{r})}$$

where $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(\vec{r}')}{\mathscr{r}} dV'$ is the magnetic vector potential, we will come back to it later.

Since $\vec{B}(\vec{r}) = \nabla \times \vec{A}(\vec{r})$, we can immediately calculate the divergence of the B field:

$$\boxed{\nabla \cdot \vec{B} = \nabla \cdot (\nabla \times \vec{A}) = 0}$$

This is Gauss's Law for magnetism, which states that there are no magnetic monopoles. By applying the divergence theorem, we can convert it into the integral form:

$$\boxed{\oint_S \vec{B} \cdot d\vec{a} = 0}$$

This is Gauss's Law for magnetism in the integral form, which implies that the net magnetic flux through any closed surface is zero. Since no magnetic monopoles have been found, magnetic field lines always form closed loops.

Starting from the expression $\vec{B}(\vec{r}) = \nabla \times \vec{A}(\vec{r})$, we can also reach a key conclusion about the curl of the magnetostatic field. Taking the curl of the equation on both sides, we get

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A},$$

now we examine the two terms on the right-hand side of the above equation one by one. For the first term, $\nabla \cdot \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \nabla \cdot \frac{\vec{J}(\vec{r}\,')}{\mathscr{r}} dV'$, according to the product rule, the integrand $\nabla \cdot \frac{\vec{J}(\vec{r}\,')}{\mathscr{r}} = \frac{(\nabla \cdot \vec{J}(\vec{r}\,'))}{\mathscr{r}} + \vec{J}(\vec{r}\,') \cdot \left(\nabla \frac{1}{\mathscr{r}}\right)$, note that ∇ acts on $\vec{r}$ rather than $\vec{r}\,'$, leading to $\nabla \cdot \vec{J}(\vec{r}\,') \equiv 0$, and accordingly, $\nabla \cdot \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \vec{J}(\vec{r}\,') \cdot \nabla \frac{1}{\mathscr{r}} dV'$. We now replace ∇ with ∇' at the expense of a minus sign

$$\nabla \cdot \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \vec{J}(\vec{r}\,') \cdot \nabla \frac{1}{\mathscr{r}} dV' = -\frac{\mu_0}{4\pi} \int \vec{J}(\vec{r}\,') \cdot \nabla' \frac{1}{\mathscr{r}} dV'$$

and apply the product rule one more time to get

$$\nabla \cdot \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{1}{\mathscr{r}} \underbrace{\nabla' \cdot \vec{J}(\vec{r}\,')}_{\substack{=0 \text{ because of} \\ \text{steady current}}} dV' - \underbrace{\frac{\mu_0}{4\pi} \oint_S \frac{\vec{J}(\vec{r}\,')}{\mathscr{r}} \cdot d\vec{a}}_{\substack{\text{can make V large enough} \\ \text{such that } \vec{J} \text{ vanishes on S}}}$$

Because we show that the two terms on the right-hand side of the above equation are both zero, we conclude that $\nabla \cdot \vec{A} = 0$, and apparently, $\nabla(\nabla \cdot \vec{A}) = 0$. We have reached the following equation: $\nabla \times \vec{B} = -\nabla^2 \vec{A}$. In order to evaluate the curl of the magnetostatic field, what left for us is to analyze the Laplacian of the vector potential. Again, we keep in mind that ∇^2 doesn't act on $\vec{r}\,'$, so

$$\nabla^2 \vec{A}(\vec{r}) = -\frac{\mu_0}{4\pi} \int \vec{J}(\vec{r}\,') \cdot \nabla^2 \frac{1}{\mathscr{r}} dV'.$$

Because $\nabla^2 \frac{1}{\mathscr{r}} = -4\pi\delta^3(\vec{\mathscr{r}})$, we obtain $\nabla^2 \vec{A}(\vec{r}) = -\frac{\mu_0}{4\pi} \int \vec{J}(\vec{r}\,') 4\pi\delta^3(\vec{r} - \vec{r}\,') dV' = -\mu_0 \vec{J}(\vec{r})$, you may have noticed that this is the magnetic counterpart of Poisson's equation, although it consists of three independent scalar equations. Now we reached another milestone:

$$\boxed{\nabla \times \vec{B}(\vec{r}) = \mu_0 \vec{J}(\vec{r})}.$$

This is known as <u>Ampere's Law in differential form</u>. It states that the curl of the magnetostatic field is equal to the product of the vacuum permeability and the local current density. This equation hold only under magnetostatic conditions; an additional term will be introduced to the right-hand side when considering a time-varying electric field.

Integrating both sides over an open surface, $\int_S \nabla \times \vec{B}(\vec{r}) \cdot d\vec{a} = \mu_0 \int_S \vec{J} \cdot d\vec{a}$, and applying Stokes' theorem, we obtain <u>Ampere's Law in integral form:</u>

$$\boxed{\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}}$$

which states that the line integral of the magnetostatic field around a closed loop is equal to the product of the vacuum permeability and the total current enclosed by the loop.

Note:

The shape of the loop and the open surface attached to it can be complex. For instance, the irregular open surface shown in the left figure below does not affect the final result of the line integral. Additionally, the loop in the right figure below is quite unusual, wrapping around the current multiple times. Although visualizing a surface attached to such a convoluted loop may be challenging, one can imagine the current piercing through this intricate surface N times, where N represents the number of turns, so $I_{enc} = NI$, where I is the orignal current. In this case, $\oint_{crazy\ loop} \vec{B} \cdot d\vec{l} = \mu_0 NI$.

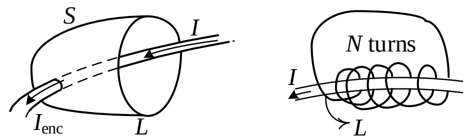

Figure. In Ampere's Law, the loop (L) and the open surface (S) can be arbitrarily complex.

If the loop encloses no current, as shown in the figure below, the line integral of the magnetostatic field around the loop is zero. In fact, within any simply-connected region free of current, the magnetostatic field is conservative, as it satisfies $\nabla \times \vec{B} = 0$.

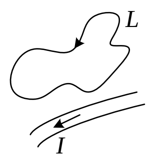

Figure. A loop emcompassing no current.

The following figure summarizes the main conclusions we reached so far.

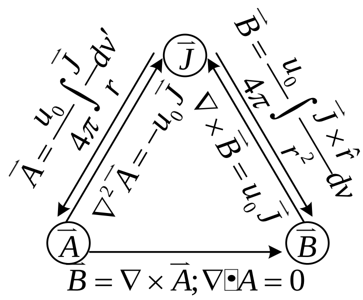

Figure. Relations between $\vec{J}$, $\vec{A}$ and $\vec{B}$.